

Assignment 1: Wound healing and time perception

Start Assignment

- Due Saturday by 1am
- Points 20
- Submitting a text entry box or a file upload
- File Types pdf
- Available Sep 13 at 12am - Sep 28 at 11:59pm

This assignment concerns a real exchange between behavioral scientists at Harvard and the statisticians Gelman and Brown, centered on a paper about wound healing. You'll start by reading Gelman's blog post, then excerpts from the original paper and from Gelman and Brown's response.

Some of this will use terminology or ideas you haven't seen yet. Don't try to look everything up or get it all on the first pass. Instead, write **down any term or idea that doesn't make sense to you yet.** Part of the point of this assignment is figuring out where the gaps in your own understanding are. We will come back to this same paper later in the course once we developed more of the background. For now focus on the high level questions, such as: what does the original study claim? Why are the authors skeptical, and what do they suggest as a remedy?

Read, in this order:

1. Andrew Gelman's blog post, [7 steps to junk science that can achieve worldly success](https://statmodeling.stat.columbia.edu/2025/01/17/7steps/)  (<https://statmodeling.stat.columbia.edu/2025/01/17/7steps/>). Read the whole post. Also read the comment by **Nick, posted January 18, 2025 at 2:48 pm**, and Andrew's reply to it right below.
2. The **abstract only** (stop there) of the original study: [Physical healing as a function of perceived time](https://www.nature.com/articles/s41598-023-50009-3)  (<https://www.nature.com/articles/s41598-023-50009-3>), Aungle & Langer, *Scientific Reports* (2023).
3. Gelman & Brown's response paper, [How statistical challenges and misreadings of the literature combine to produce unreplicable science: An example from psychology](https://sites.stat.columbia.edu/gelman/research/published/healing3.pdf)  (<https://sites.stat.columbia.edu/gelman/research/published/healing3.pdf>) — **read through the end of Section 1** (i.e. through Section 1.2, "A questionable finding in psychology"). You can stop before Section 2.
4. From that same Gelman & Brown paper, also read **Section 4.2, "Misinterpretation of the literature and relevance to the replication crisis"** (p. 12). You can skip everything in between (Sections 2, 3, and the rest of Section 4).
5. It is worth noting that the authors of the original paper have [responded to Gelman & Brown](https://journals.sagepub.com/doi/pdf/10.1177/25152459261417257)  (<https://journals.sagepub.com/doi/pdf/10.1177/25152459261417257>) (Aungle, Chen, & Holmes). Read only the section **"The Literature Reviewed by Gelman and Brown."** Their response also addresses the concerns about the statistical model from Section 2 of Gelman & Brown's paper —

that part is beyond the scope of this assignment, and I'll leave open here whether it adequately addresses those concerns.

Part (i) — The garden of forking paths

The point of this problem is to make the idea of the garden of forking paths more quantitative. Suppose there are N different reasonable ways we could analyze a given dataset — different choices of which variables to include, which distribution to assume, even how to phrase the question. Suppose the hypothesis H we're testing is actually **true** (e.g. there's no real effect of variable X on Y), but *each* of the N analysis choices has some probability q of leading us to (incorrectly) reject H anyway. q is sometimes called the **Type I error rate**.

(a) If we try all N ways of analyzing the data and assume the N outcomes are **independent** (of course, this is a simplification — in reality they wouldn't be), what is the probability that *at least one* of the N analyses leads us to (incorrectly) reject H ?

(b) Suppose $q = 0.05$. How many analyses N would it take for this probability to reach 50%? Test your answer with simulations.

(c) How do you think the answer would change if the analyses are not independent? Test your answer with simulations.

Part (ii) — Exploring the real healing data

In the actual study, 33 participants had cupping marks (small bruises) induced, and healing was later assessed from photographs by 25 external raters on a 0–10 scale (0 = not at all healed, 10 = completely healed). Each participant went through this three times — three separate sessions, each on a different occasion — under three different "perceived time" conditions (told the recovery interval was 14, 28, or 56 minutes even though it was actually 28 minutes in each case).

You can load the data with the following code. The last line turns the dataframe into a numpy 2d array.

```
import numpy as np
import pandas as pd

url = (
    "https://media.springernature.com/original/springer-static/esm/"
    "art%3A10.1038%2Fs41598-023-50009-3/MediaObjects/"
    "41598_2023_50009_MOESM4_ESM.csv"
)
df = pd.read_csv(url)

df["S"] = pd.factorize(df["Subject"])[0] + 1    # subject id, 1-33
df["R"] = pd.factorize(df["ResponseId"])[0] + 1 # rater id, 1-25

X = df[["Condition", "Session", "Healing", "Age", "Gender", "S", "R"]].to_numpy()
X
```

Each row of **X** is a single healing rating — one rater's score for one participant's photo from one session. The columns are:

- **Condition** — the perceived-time condition for that session (14, 28, or 56)
- **Session** — which of the participant's three visits this was (1st, 2nd, or 3rd, in chronological order)
- **Healing** — the 0–10 healing rating given by that rater
- **Age**, **Gender** — characteristics of the participant, repeated across all of their rows
- **S** — a participant id
- **R** — a rater id

Examine the data in Python to answer the following:

- Were raters assigned to particular sessions, or could a rater see photos from more than one session?
- Did all participants get rated by the same set of raters? Did some raters end up rating certain conditions more than others?
- Comment on these design choices — how do you think they might influence the results?

Basic analysis.

- Compute the average healing rating in each condition.
- Now compute the average healing rating in each condition, separately for each session.
- Which is bigger: the variance in healing ratings *within* a condition (across participants), or the variance *across* condition means? What does that comparison tell you?

Part (iii) — A randomization test

In the real data, compute the difference in mean healing rating between the 14-minute and 56-minute conditions — the two conditions the paper contrasts most starkly.

Now suppose condition actually had no effect on healing at all. We can simulate this "null hypothesis" that every healing rating was really an independent draw from one overall pool of healing scores by replacing every value in the **Healing** column with a value drawn at random (with replacement) from the full pool of healing ratings in the dataset. The sampling ignores which participant, session, or condition the healing value came from. With this new data, recompute the 14-minute vs. 56-minute difference in this randomized version of the data.

Repeat that resampling a few hundred times, recording the difference each time. What fraction of these random differences are at least as large, in absolute value, as the difference you actually observed in the real data? That fraction estimates the probability of seeing a difference this big by chance alone, if condition truly had no effect — connect this back to the Type I error idea from Part (i).

Part (iv) — Reacting to a disagreement between experts

You've now seen both sides of a real, ongoing disagreement. Write a short paragraph reacting to this disagreement:

- Are you persuaded by one side more than the other? Why or why not?
- Do you feel qualified to evaluate the research yet? What would you need to know, or be able to do, to feel more confident in your evaluation?
- How might LLM tools help or hurt you in trying to navigate a disagreement like this? For example, could an LLM help you check whether a citation actually supports the claim being made of it? Could it also make it easier to be misled?
- Did doing your own analysis in Parts (ii) and (iii) change your view at all? Why or why not?
- What role should institutions such as journals, universities, funding agencies, professional societies play in encouraging good statistical practice and discouraging garden-of-forking-paths-style analysis? Do you think the incentives those institutions currently create point the right way?

Optional: if you'd like to go one step further, Gelman and Brown have in turn responded to Aungle et al.'s response: [Revision of Reply to Aungle et al.](#) 

https://sites.stat.columbia.edu/gelman/research/published/Revision_of_Reply_to_Aungle_et_al.pdf This is not required reading for this assignment.